

Israel C. Ribeiro, Ph.D.

Materials Science | R&D Data Scientist | Computational Chemistry

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Professional Summary

Ph.D. computational chemist and data scientist with 5+ years accelerating materials discovery through first-principles simulation, high-throughput screening, and machine-learning workflows. Currently applying neural network potentials and ML-driven structural search in an international EU consortium at UMONS (Belgium). Proven track record bridging theory and experiment — from DFT design to sol-gel synthesis — with peer-reviewed papers in several scientific journals. MBA in Data Science & Analytics and B.Sc. in Materials Chemistry.

Education

Ph.D., Physical Chemistry — Computational Materials Science

University of São Paulo (USP)

Advisor: Prof. Juarez L. F. Da Silva

2020 – 2025

São Carlos, Brazil

MBA, Data Science & Analytics

University of São Paulo (USP)

Advisor: Prof. Gabrielle Lombardi

2024 – 2026

Online

B.Sc., Chemistry (Minor: Materials Chemistry)

Federal University of São João del Rei (UFSJ)

Advisor: Prof. Horacio Wagner

2015 – 2020

São João del Rei, Brazil

Technical Skills

- Computational Materials Design: DFT / first-principles (VASP, Quantum ESPRESSO, FHI-aims, Gaussian); high-throughput screening; surface and interface modelling; neural network interatomic potentials (NNIPs); structural search algorithms; tight-binding frameworks; multi-scale simulation.
- Data Science & Machine Learning: k-means and hierarchical clustering, regression, classification, PCA and dimensionality reduction; scikit-learn, pandas, NumPy, SciPy; materials descriptor engineering; large-scale data pipelines in Python/Bash; Matplotlib, Seaborn, Plotly.
- HPC & Dev Tools: Linux, Slurm, Git, Jupyter, LaTeX; Python GUI development; end-to-end workflow automation.
- Experimental Characterisation: sol-gel synthesis; XRD, FTIR, UV-Vis, TGA, BET, SEM/TEM, ICP; bridging computational predictions and synthesis targets.

Industry Experience

Chemical Process Intern

AMG Brazil

2015 – 2018

São João del Rei, Brazil

- Executed fluoride/chloride and ICP elemental analyses, feeding data into statistical quality-control dashboards to drive compliance decisions.
- Led process-optimisation trials and authored technical reports adopted by production management; spearheaded 5S programme that measurably improved laboratory efficiency.
- Ensured product conformance with industry regulations through systematic performance testing and root-cause documentation.

Chemical Technician

Resind Ltda.

2019 – 2020

São João del Rei, Brazil

- Managed quality assurance for primary chemical products using X-ray quantitative analysis, maintaining traceability and regulatory standards compliance across production batches.

Research Experience

Postdoctoral Researcher – Computational Materials Science

Université de Mons (UMONS)

Jan 2026 – Present

Mons, Belgium

- Built Python GUIs and automation pipelines for machine-learning interatomic potential (MLIP) workflows, cutting manual setup time across a multi-partner EU consortium.
- Deployed neural network potentials coupled with structural search algorithms to model hybrid organic-inorganic interfaces and quantify band alignment mechanisms.
- Engineered HPC data pipelines (Slurm) to process MD trajectories and compute electronic properties via tight-binding frameworks at scale.
- Coordinated deliverables across theoretical and experimental partners, translating simulation outputs into actionable insights for materials synthesis teams.

Doctoral Researcher – Computational Materials Science

University of São Paulo (USP)

Nov 2020 – Sep 2025

São Carlos, Brazil

- Designed and executed DFT / first-principles studies on 2D perovskite thin-film passivation, resolving structural, energetic, and optoelectronic properties at the atomic scale.
- Applied k-means and hierarchical clustering (scikit-learn) to screen organic cation candidates, shrinking the experimental search space and prioritising top materials for synthesis.
- Automated end-to-end simulation workflows in Python/Bash, improving reproducibility and reducing compute turnaround on HPC clusters.
- Delivered scientific publications (6 peer-reviewed) in ACS Omega, ACS Applied Energy Materials, and Applied Surface Science; communicated results to both theorists and experimentalists.

Undergraduate Researcher – Materials Chemistry

Federal University of São João del Rei (UFSJ)

2017 – 2020

São João del Rei, Brazil

- Synthesised rare-earth-doped inorganic nanomaterials via sol-gel and characterised them by XRD, FTIR, UV-Vis, TGA, BET, and SEM/TEM.
- Performed first-principles electronic-structure calculations, building an integrated theory-experiment workflow.

Selected Publications

- Ribeiro, I.C. et al. "Combining DFT and Clustering to Screen Organic Cations for Halide Perovskite Solar Cells." *ACS Omega*, 2026.
- Ribeiro, I.C. et al. "Phosphonic-Acid Passivation for Surface Termination of Hybrid Halide Perovskites." *ACS Omega*, 2026.
- Ribeiro, I.C. et al. "Impact of Thin-Film Thickness on 2D Perovskite Optoelectronic Properties." *ACS Applied Energy Materials*, 2025.
- Ribeiro, I.C. et al. "Organic-Cation Passivation in 2D Perovskite Thin Films." *Applied Surface Science*, 2024.

Full list: [Google Scholar](#) (10+ publications).

Languages

Portuguese (Native) | English (C1) | French (B1) | Spanish (A1)